Research Protocol

Objective: The goal of this study is to build a predictive model of NFL team performance using longitudinal multilevel data. A historical dataset will be used to examine how performance, coaching and team variables predicts winning in the NFL. A forecast of team wins for the 2025 season will be done to compare the model's predictions to actual outcomes. The predictive model will also be compared in terms of RMSE to published approaches in the sports analytics literature.

Data Structure: The data is obtained from the nflreadr and nflfastR R packages, which provide historical information on NFL teams (Seasons from 2002, 1st season in which the NFL operated with the current 32 team structure, to the 2024 season will be used). After cleaning the data, the dataset will consist of one row per team per season. Because NFL team appears repeatedly across multiple years, the data has a multilevel structure in which teams act as clusters and seasons are repeated measurements nested within each team. Thus, mixed-effects models are required to account for within-team correlation and the dependence of observations across seasons.

Variables: The outcome variable is measured either as regular season win percentage or regular season win total. The predictor variables include performance metrics such as offensive EPA, defensive EPA, and special teams EPA, each providing an efficiency-based measure of on-field performance. Also, coaching stability metrics are included such as years as head coach, tenure with current team, and years with offensive and defensive coordinators. Lastly, team information is included such as average player age and experience, and number of rookies. Correlation analysis, variance inflation factors, and model selection criteria such as AIC will guide the final inclusion of predictors.

Model Specification: Depending on the chosen outcome and the results of model diagnostics, team performance will be modeled using one of several mixed-effects frameworks. Raw win counts will be modeled with a Poisson generalized linear mixed model and a Negative Binomial generalized linear mixed model. Win percentage will be modeled with a linear mixed model. Each model will include a random intercept for teams, capturing team differences in baseline performance and a random slope for season, allowing each team to follow its own trajectory over time. Comparison of multiple model types of linear mixed models, Poisson GLMMs, and Negative Binomial GLMMs will be done to identify the best specification within each model class using AIC. After selecting the top model from each class, the finalists of each class will be compared using RMSE and the model that achieves the best predictive performance will be selected.

Prediction Component: Once the model is trained using historical data from 2002 through 2024, it will be used to generate predicted wins for the 2025 season and compared to actual season results.

Limitations and Bias: There are a few limitations in this type of analysis. Firstly, the dataset does not include qualitative factors such as team morale or locker-room dynamics. Moreover, the model cannot anticipate mid-season events such as injuries, trades and coaching changes. In short, there are a variety of factors that affect winning that cannot be measured and thus, the omission of those variables in the model will undoubtedly affect its predictive powers. Despite these challenges, the data available should be enough to create a relatively robust predictive model that can be used to forecast future win totals for teams and can guide franchise level decisions that promote winning in the NFL.